

# Peixuan Yang

*Incoming BEng (Hons) Student in Computer Science and Electronic Engineering*

Suzhou, China; relocating to Liverpool, UK (Sep 2026) | +86 189 4294 3381 | Peixuan.Yang24@student.xjtlu.edu.cn

**Research Interests:** AI hardware-world-model co-design; photonic computing and computer architecture; efficient generative inference; trustworthy autonomous and multi-robot systems.

## EDUCATION

**University of Liverpool | Liverpool, United Kingdom | Sep 2026 - Jun 2028 (expected)**

*BEng (Hons), Computer Science and Electronic Engineering (Years 2-3 in Liverpool)*

**Xi'an Jiaotong-Liverpool University (XJTLU) | Suzhou, China | Sep 2024 - Jul 2026**

*University of Liverpool International Education Programme; programme title at XJTLU: Computer Science and Technology*

- 2025/26 weighted average: 71% (First-Class level). Selected marks: Computer Systems 88%, Engineering Mathematics II 86%, Digital Electronics I 76%, and Data Structures and Algorithms 76%.

**CEAIE Innovative Talents Science Training Programme | Mar 2025 - May 2025**

- Machine Learning, Network Security and Large Language Models, instructed by Prof. Pietro Liò (University of Cambridge); final grade 97.40 / A+.

## RESEARCH EXPERIENCE

**PhotonWM: Trace-Driven MRR Acceleration for Autoregressive World Models | Jul 2026 - Present**

*Student Researcher | Supervisor: Prof. Jiaqi Liu*

- Co-designed a hybrid photonic-electronic inference architecture that maps fixed-weight Linear operators to 64 x 64 microring-resonator arrays while retaining dynamic attention, nonlinearities, sampling and control on the electronic path.
- Reproduced and profiled the official LPWM BAIR-128 checkpoint on an RTX 5090; constructed operator and tensor-shape ledgers for a 30-step rollout and identified recurrent Dynamics execution as 91.56% of measured stage latency.
- Integrated measured traces with LLMCompass digital baselines, SimPhony device/circuit costs and an event-driven scheduler for weight residency, MRR programming, conversion, latency, energy and recurrent-quality analysis; contributed the Evaluation methodology and bilingual manuscript revision. Full paper submitted to DATE 2027; decision pending.

**SURF 2026: Collaborative Proof-of-Sensing and Trusted Multi-Robot Auditing | Jun 2026 - Aug 2026**

*Research Student, Blockchain Security Group | Supervisor: Prof. Limin Yu*

- Conducted largely independent literature-to-prototype research on Collaborative Proof-of-Sensing (CPoS) and lightweight trusted auditing for UAV, surface and underwater robot missions.
- Implemented and tested a Hardhat local EVM network, Solidity sensing registry and ethers.js client for digest submission, retrieval and audit events; documented reproducible transaction and gas evidence while keeping raw sensing data off-chain.
- Built a ROS 2 Jazzy, Gazebo Harmonic and PX4 v1.17 three-X500 environment with independent identities and namespaces, and validated a headless take-off, hover, waypoint, return and landing workflow. Specified the next-stage protocol for signed multi-source mission evidence and future end-to-end physical-sensing verification.

**Low-Resource Cantonese-English Neural Machine Translation | Jun 2025 - Jan 2026**

*Researcher | Co-second Author and Presenter, ICRAI 2025*

- Constructed and cleaned a 1.1-million-pair Cantonese-English corpus through syntactic filtering, normalisation, deduplication and LaBSE-based semantic alignment.
- Fine-tuned NLLB-200-distilled-600M and designed a nested pseudo-data sampling strategy; the optimal 200k:200k pseudo-to-real setting achieved BLEU 29.63 and chrF 56.65.

## SELECTED PUBLICATION

Y. Wang, Y. Gao, P. Yang, and B. Zhao, "Optimized Fine-tuning and Pseudo-Data Strategies for Cross-Domain Low-Resource Language Cantonese-English Neural Machine Translation," 11th International Conference on Robotics and Artificial Intelligence (ICRAI), Nagoya, Japan, 2025.

- Co-second author and presenter. Built and cleaned a 1.1-million-pair Cantonese-English corpus, contributed full-parameter NLLB-200-600M fine-tuning and nested pseudo-data sampling; the best in-domain setting achieved BLEU 29.63 and chrF 56.65.

## PROFESSIONAL EXPERIENCE

---

### Research Institute of Tsinghua University in Shenzhen (RITS) | Shenzhen, China | May 2026 - Jul 2026

*Research and Development Intern, Marine Hydrogen Energy R&D Center*

- Completed a two-month R&D placement in the Marine Hydrogen Energy R&D Center; commended in the official internship certificate for initiative, collaboration and strong overall performance.

### Wuhan University City Hospital Co., Ltd. | Wuhan, China | Jan 2026 - Mar 2026

*Medical Data Analysis / IT R&D Intern*

- Applied Python to process and analyse operational data, supported technical troubleshooting, and contributed to improving internal healthcare IT workflows.
- Evaluated AI-assisted troubleshooting methods and helped organise internal knowledge-base content for more consistent technical support.

### Wuhan Dongfangyunjin Information Technology Co., Ltd. | Wuhan, China | Feb 2023 - Dec 2023

*Backend and UI Programmer (Medical Devices)*

- Developed web interfaces and external API integrations supporting secure data exchange and collection of partner bidding information.
- Implemented an RBAC user-management module enabling administrators to perform controlled permission and account CRUD operations.

## SELECTED TECHNICAL PROJECT

---

### HiAgent Intelligent HR System | XJTLU Hackathon

*Lead Knowledge Base Engineer*

- Designed a retrieval-augmented candidate-evaluation workflow, curated and tagged job descriptions, resumes and interview records for optimised vector retrieval and LLM interpretation.
- Integrated PDF/Word parsing and orchestrated an end-to-end workflow linking job-description analysis, resume parsing, candidate scoring and dynamic interview-question generation.

## TECHNICAL SKILLS

---

**Research and ML:** Python, PyTorch, model profiling, workload tracing, reproducible experimentation, NLLB/LLM fine-tuning.

**AI systems and hardware:** LPWM and world models, LLMCompass, SimPhony, microring-resonator photonic computing, hybrid photonic-electronic architectures.

**Robotics and security:** ROS 2, PX4 SITL, Gazebo, Solidity, Hardhat, ethers.js, blockchain and permissioned-ledger prototyping.

**Software:** Git, Linux, Java, SQL, REST APIs, LaTeX.

## SELECTED HONOURS

---

- Best Presentation Award - ICRAI 2025, Nagoya, Japan.
- Jiangsu Province Anti-Drug Promotion Ambassador and Outstanding Anti-Drug Youth.